

YIFAN XU

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Research interests: GPU-accelerated optimization – algorithm design and high-performance implementation for large-scale linear and nonlinear programs – and optimal-transport methods for graph alignment, with applications across machine learning and sciences.

EDUCATION

AUG 2023 - MAY 2027	RICE UNIVERSITY B.S. in OPERATIONS RESEARCH, B.A. in COMPUTER SCIENCE AND MATHEMATICS (GPA: 3.91/4.00) Relevant Coursework: <i>Convex Optimization, Linear & Integer Programming, Optimization Foundations of Data Science, Distributed Optimization & ML, Deep Learning Systems, Computer Systems, Concurrent Program Design, Honors Linear Algebra, Probability Theory, Algorithms, Complexity & Approximation</i>
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PUBLICATIONS

Name in **bold**.

- [1] **Yifan Xu**, Carlos Taveras, Lydia Beaudrot, and César A. Uribe. Identifying Common Backbones of Interactions Underlying Food Webs via Non-Deterministic Alignments. In *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5961–5965, 2026
- [2] **Yifan Xu**, Carlos A. Taveras, Lydia Beaudrot, and César A. Uribe. Wasserstein Motifs: Non-deterministic Alignment of Ecological Networks. Accepted as an Oral Presentation at the Learning Meaningful Representations of Life (LMRL) Workshop at ICLR 2026 (non-archival); manuscript in preparation for the SIAM Journal on Life Sciences
- [3] Sebin J. Gracy, **Yifan Xu**, Ji Liu, Tamer Başar, and César A. Uribe. Networked Competitive Bivirus SIS Model: Analysis of the Discrete-Time Case. Submitted to *IEEE Open Journal of Control Systems*
- [4] **Yifan Xu** and Aung Aung Phyto Wai. Evaluation of Steady-State Visual Evoked Potentials (SSVEP) Stimuli Design for Visual Field Assessment. In *2022 International Conference on Cyberworlds (CW)*, pages 179–186, 2022

POSTERS & ORAL TALKS

- *Wasserstein Motifs: Non-deterministic Alignment of Ecological Networks*. Presented at:
 - 2026 IEEE International Conference on Acoustics, Speech, and Signal Processing, May 2026, Barcelona, Spain. [Poster](#)
 - 2026 INFORMS Optimization Society Conference, March 2026, Atlanta, GA. **Oral Presentation**. [Slides](#)
- *Enabling Workforce Intelligence through Occupational Taxonomy Alignment*. Presented at:
 - 12th International Conference on Computational Social Science (IC2S2), July 2026, Burlington VT. [Poster](#)
 - 15th Workshop on Economic & Financial Networks @ NetSci 2026, June 2026, Boston, MA. **Oral Presentation**. [Slides](#)
 - 11th Annual Conference on Network Science and Economics, April 2026, Miami, FL. [Poster](#)
- *Numerical Analysis of Discrete-time Networked Competitive Bivirus SIS Models*. Talk at the Gulf Coast Undergraduate Research Symposium 2024, Houston, TX. **Oral Presentation**. [Slides](#)

SELECTED RESEARCH EXPERIENCES

MAY 2026 - PRESENT	MASSACHUSETTS INSTITUTE OF TECHNOLOGY Kernel-agnostic Method of Multipliers for GPU-Accelerated Nonlinear Programming <i>Research Intern</i> <u>Advisor:</u> Sungho Shin • Investigating GPU-friendly augmented-Lagrangian / method-of-multipliers algorithms for large-scale constrained NLPs, analyzing how the penalty-kernel choice shapes the conditioning and GPU-factorizability of the KKT system. • Prototyping a mixed-kernel formulation (NCL quadratic equality kernel + exponential inequality kernel) to preserve a GPU-friendly KKT structure while collapsing inequality handling into a single equality-constrained Newton solve.
OCT 2025 - PRESENT	RICE UNIVERSITY & JOHNS HOPKINS UNIVERSITY GPU-accelerated Bregman Douglas-Rachford Splitting for Discrete Optimal Transport <i>Undergraduate Researcher</i> <u>Advisor:</u> Shiqian Ma • Developing a GPU-accelerated solver for large-scale Discrete Optimal Transport via Bregman Douglas–Rachford Splitting, using entropy/Bregman geometry for numerically stable iterates. • Derived a factored log-domain representation and exploited cost separability to cut per-iteration complexity from $O(N^2)$ to $O(N^{3/2})$ and memory from 550 GB to 20 MB at 512×512. • Designed a c-transform dual post-processing step producing by-construction feasible dual certificates, cutting iterations to a 10^{-4} KKT residual by $\sim 82\%$ on DOTmark and settling all instances faster than state-of-the-art GPU OT solvers; implemented with fused CUDA kernels and CUDA Graphs.

AUG 2025 - PRESENT	RICE UNIVERSITY & MASSACHUSETTS INSTITUTE OF TECHNOLOGY
	Enabling Workforce Intelligence through Occupational Taxonomy Alignment <i>Undergraduate Researcher</i> <u>Advisors:</u> César A. Uribe, Isabella Loaiza - Built a task meta-network from O*NET data (2017–2024) using embedding-based clustering and graph-lasso-estimated task dependencies. - Formulated occupational alignment as a Fused Gromov–Wasserstein problem integrating task semantics and network structure for soft, interpretable matching. - Scaled the approach to around 900K occupation pairs, achieving 80% agreement with official crosswalks while exposing task-driven sources of misalignment.
MAY 2024 - PRESENT	RICE UNIVERSITY & MICHIGAN STATE UNIVERSITY
	Wasserstein Motif: Optimal Transport for Ecological Network Alignment <i>Undergraduate Researcher</i> <u>Advisors:</u> César A. Uribe, Lydia Beaudrot Developed the first mathematically rigorous framework for ecological network alignment using network motifs and optimal transport. Designed a provably convergent variant of the KL-BAPG algorithm with Dykstra projections to compute many-to-many correspondences. Demonstrated a 158x computational speedup and superior ecological interpretability across 129 Sub-Saharan African food webs.

HONORS & AWARDS

- Rice Undergraduate Scholar Program. Academic Year 2026-2027.
- School of Engineering Outstanding Junior Award (one of nine nominees). Academic Year 2025-2026.
- Honorable Mention, *2026 Computing Research Association Outstanding Undergraduate Researcher*.
- Cornell, Maryland, Max Planck Pre-Doctoral Research School (2x). April 2025, April 2026.
- Runner up, Poster competition at *Rice School of Engineering Undergraduate Research Symposium* (\$1,000). Feb 2025.
- Bishnoi SER Scholarship and CREST Scholarship (\$3,000). Sept 2024.
- President's Honor Roll. Fall 2023, Fall 2024, Spring 2025.

TEACHING & SERVICE

JAN 2026 - MAY 2026	Optimization Mentor for DecisionLab
	<i>The Official Student Chapter of INFORMS at Rice University</i> Mentored 5 undergraduates across a 12-week applied OR program on modelling the final exam scheduling problem at Rice with IPs/MIPs; created and delivered mini-lectures on LPs, IPs, and mathematical modeling. Slides
FEB 2025	Workshop Creator & Instructor <i>Google/Rice Data Science and Sustainability Program</i> Co-designed and delivered a curriculum on data science applications in ecology and sustainability, taught in person to community-college students from Lone Star, San Jacinto, and Houston Community College. Colab
AUG 2024 - DEC 2025	Teaching Assistant Algorithms, Linear Algebra, and Proof Writing (2x) <i>Rice Department of Computer Science and Department of Mathematics</i> Supported 50+ students weekly in linear algebra, proof writing, and algorithms via office hours; as head TA, coordinated fellow TAs and authored grading rubrics for consistency.

SOFTWARE & PERSONAL PROJECTS

OCT 2025 - DEC 2025	M3ssag1n8: a Web-based Messaging Application
	Built a real-time messaging web application on an MVC architecture in TypeScript, with private workspaces/channels and SSE connections for real-time synchronization. Architected the model layer — AuthModel for authentication and DatabaseModel for CRUD operations — with integrated SSE event handling for live updates. Implemented concurrency control via a custom <code>Limit<T></code> class with promise chaining to prevent race conditions during rapid post creation and reaction changes.
AUG 2025 - OCT 2025	OwIDB: a Network Accessible NoSQL Document Database Built a Go REST API for a NoSQL JSON document store with schema validation, token-based authentication, and SSE subscriptions. Wrote a skip-list implementation of database lookup for concurrency and efficiency. Implemented atomic writes, command line configuration, slog logging, and graceful shutdown with tests.

SKILLS

Languages:	Python, CUDA, Julia, C, Java, GDScript, R, MATLAB, Golang, JavaScript, Haskell
Packages:	PyTorch, JAX, NumPy, GurobiPy, NetworkX, Pandas, scikit-learn, PsychoPy, Matplotlib
Math:	Optimal Transport, Convex Optimization, Numerical Optimization, Spectral Graph Theory